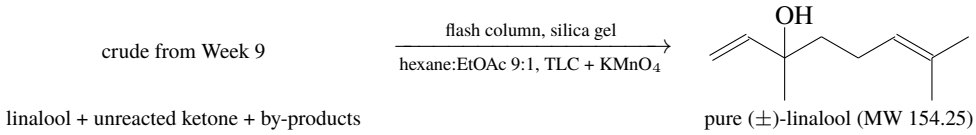
ORGANIC EXPERIMENT 4, PART 2 — PRE-LAB CHECKLIST

Flash Column Chromatography of Crude (±)-Linalool

Condensed from the CM3191 Organic Lab Manual AY26/27 (Dr Hoang T. G., pp. 24–27) and lecture CM3191 Org L5.

Week 11, **Thursday 29 October 2026**, Thursday lab, **Group A**. Continues from Part 1 (Week 9, 15 October): you start with the crude linalool you made then.



1. Numbers you need today

Item	Value	Note
Theoretical yield of linalool	**0.460 g**	from 0.44 mL ketone (2.98 mmol); with your actual volume V: $1.045\,V$ g
Crude mass (Part 1)	_____ g	copy from your Week-9 datasheet
Column eluent	200 mL	hexane:EtOAc 9:1 = **180 mL hexane + 20 mL EtOAc**
Silica gel	∼4 spoons	as a slurry in 20–30 mL eluent; settled bed **12–15 cm**
Sand on top	∼0.5 cm	protects the silica surface
Target R_f of product	≈0.3	L5 rule of thumb; check your Week-9 TLC (9:1)

2. Bring / prepare before the session

☐ **Crude linalool** flask from Week 9
☐ Official datasheet (manual p. 27): **purified product** section is filled today **[CONFIRM: same signed sheet as Week 9, or a new copy]**
☐ Ruler + pencil (TLC distances in cm)
☐ Lab coat, goggles, nitrile gloves, covered shoes
☐ Your **Week-9 TLC result** (ketone vs. product R_f, 9:1)
☐ This bench sheet
☐ Phone for photos of every balance reading
☐ Test-tube rack; marker to **number every tube**

3. Session timeline (estimate, not in the manual)

#	Stage	Approx. time
1	Mix 200 mL eluent; make silica slurry; cotton + sand in the column; check flow	15 min
2	Pour slurry, settle to 12–15 cm, drain to just above silica, add 0.5 cm sand	20 min
3	Load crude neat; rinse 1 (∼1 mL), rinse 2 (2–3 mL); absorb each	10 min
4	Add eluent, air pressure, collect fractions; **do not pause once loaded**	30–45 min
5	TLC of fractions (4–6 spots per plate, $KMnO_4$); decide which tubes to combine	30 min
6	Combine into pre-weighed rbf; rotavap (in batches); vacuum 20 min; weigh	35 min
7	IR; NMR sample; datasheet photos and GTA signature; dry and empty the column	25 min

SAFETY — from the manual (pp. 20, 24) and L5

Silica gel is toxic if inhaled: handle the dry powder only in the fume cupboard and move it between hoods only as a wet slurry. Hexane is flammable and a neurotoxin; ethyl acetate is highly flammable. Linalool is a skin and eye irritant. $KMnO_4$ stain is an oxidiser and stains skin and clothes. After the column, blow air through until the silica is dry and free-flowing, then empty it into the **silica waste** container (L5).

Added notes (not in the manual). Clamp the column vertically before applying air pressure, and use only gentle pressure. If DCM is used for loading, its waste goes to the **halogenated** bottle; hexane/EtOAc fractions you do not keep go to **non-halogenated** waste. **[CONFIRM: waste bins in the organic lab]**

BENCH MANUAL

Grignard Reaction, Part 2: Flash Column Chromatography

Tick each box as you complete the step. "Why" notes explain the purpose; they are not extra steps. Photos of the manual's column and TLC figures are in Appendix A.

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1. Preparation

☐ 1. In a **250 mL beaker**, prepare **200 mL hexane:EtOAc 9:1** (180 mL hexane + 20 mL EtOAc). Get **2 clean Pasteur pipettes**: one for the sample, one for the eluent.

☐ 2. In the fume cupboard: **20–30 mL eluent + about 4 spoons of silica gel** in a 250 mL beaker or conical flask; mix well into a slurry.

Why: wet (slurry) packing gives a bed without air pockets, and keeps silica dust out of the air.

☐ 3. Column: push in a small plug of **cotton**, add a **layer of sand**. Clamp the column **vertically** on a retort stand in the hood. Put a **100 mL conical flask** under it.

☐ 4. Add **10 mL eluent** to the column and open the stopcock to **check the flow**.

2. Packing the column

☐ 5. Pour the slurry into the column through a **solid funnel**. Rinse the beaker and the column walls with eluent; let the silica **settle**.

☐ 6. Settled silica should be **12–15 cm** long. Use the **air pump** if the flow is too slow.

☐ 7. Drain until the solvent is **just above the silica**. **Never let it fall below the silica surface.**

Why: a dry top cracks and forms channels; the sample then runs down the cracks and the separation is lost.

☐ 8. Add ∼**0.5 cm sand** on top; rinse the walls with eluent; drain again to **just above the sand/silica**.

☐ 9. The eluent collected while packing is clean: it **can be reused** for running the column.

3. Loading the sample

☐ 10. With a **long Pasteur pipette**, transfer the **crude oil** (neat) onto the column, **along the inner wall, in a circle**. Open the stopcock and let it **absorb into the silica**.

Why: a narrow, even starting band gives sharp, well-separated bands; dropping it in one spot "drills" a hole.

☐ 11. **Rinse 1**: ∼**1 mL eluent** (or **DCM** if the crude does not dissolve) into the rbf; swirl; transfer to the column; let it absorb.

☐ 12. **Rinse 2**: **2–3 mL eluent**; swirl; transfer; let it absorb.

Why: every extra mL at this stage widens the band. L5 suggests minimal DCM in general; for this experiment follow the manual (neat + eluent rinses).

4. Running the column

☐ 13. Add ∼**200 mL eluent gently** (do not disturb the silica). Apply **air pressure**. Collect in **numbered test tubes**, starting **immediately** (L5).

Why: some compounds elute quickly. Do not stop the column once loaded: bands spread by diffusion and separation and yield drop (L5).

☐ 14. TLC the tubes: **4–6 spots per plate**; include the **starting ketone** as a reference; elute 9:1; **$KMnO_4$ stain**. The spotting capillary can be washed with acetone and reused.

Why: neither ketone nor linalool is UV-active; $KMnO_4$ oxidises their C=C (yellow-brown spot on purple). The product is often dilute in each tube: spot several times on the same point.

☐ 15. If the product has **not eluted**, add more eluent or **increase the polarity** (more EtOAc). Continue until all of the product is off.

☐ 16. **Keep every tube** until TLC has shown which ones contain only the product.

5. Isolation and characterisation

☐ 17. Pre-weigh a **dry 100 or 250 mL rbf** (photo). Combine **all tubes containing only the product**.

☐ 18. Rotary-evaporate (**in batches** if needed). Then **vacuum 20 min**. Weigh (photo); fill in the **purified product** section of the datasheet.

Why: residual EtOAc/hexane adds mass; linalool is fairly volatile, so do not leave it under vacuum much longer than needed.

☐ 19. Record an **IR spectrum**; submit an **NMR sample** in $CDCl_3$.

☐ 20. Blow air through the column until the silica is **dry and free-flowing**; empty it into the **silica waste** container (L5).

☐ 21. Show photos to the GTA; get the **printed datasheet signed**.

DATASHEET — WORKING LOG

Grignard Reaction, Part 2: Flash Column Chromatography

Name: KIM DONGHWAN Fumehood: _____ Date: 29 Oct 2026 Photos taken: ☐

Working log for your own use. The **official** datasheet is the last page: print it, get it signed, submit it.

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1. Masses and yields

Ketone used, Week 9 (mL)		Theoretical $= 1.045 \times V$ (g)	
Crude mass, Week 9 (g)		Crude yield (%)	
Empty rbf, today (g)		rbf + purified product (g)	
Purified product (g)		**Purified yield (%)**	
Recovery from crude = purified / crude (%)		IR file / NMR tube no.	

2. Column record

Silica bed height (cm)		Loading: neat / DCM rinse?	
Approx. volume per tube (mL)		Total eluent used (mL)	
Polarity increased? (to what ratio)		Product in tubes no.	

3. Fraction TLC (hexane:EtOAc 9:1, $KMnO_4$) — mark K = ketone, P = product, M = mixed, – = nothing

Tube	Spots	Tube	Spots	Tube	Spots	Tube	Spots
1		6		11		16	
2		7		12		17	
3		8		13		18	
4		9		14		19	
5		10		15		20	

4. R_f values (distances in cm; copy the drawing onto the official datasheet)

Spot	Spot (cm)	Front (cm)	R_f	Note
6-Methyl-5-hepten-2-one (reference)				
Linalool (product fractions)				
Other spot(s)				

5. Checkpoints

Stage	**Expected** [source]	Observed	✓
Packed bed	12–15 cm, no cracks or bubbles [M]		
Loading	sample absorbed as a narrow band [M]		
TLC under UV	nothing visible; $KMnO_4$ needed [M]		
Order of elution	ketone before linalool (less polar first) [E]		
Purified product	colourless oil, floral smell [E]		

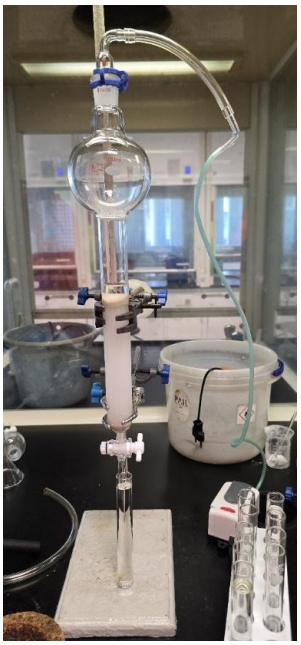
Sources: [M] lab manual; [E] general expectation, not in the manual — record what you actually see.

6. Other observations

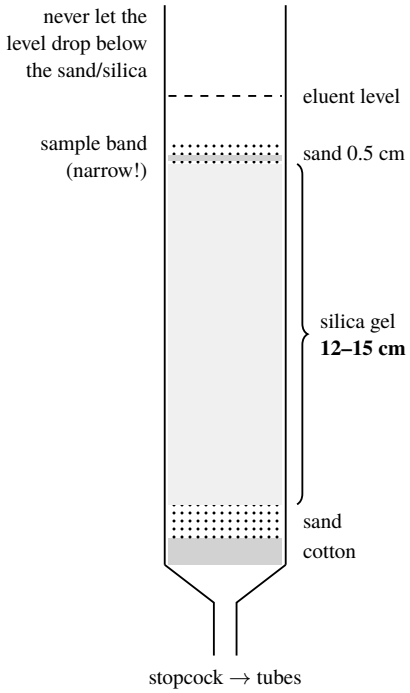
APPENDIX A — COLUMN AND TLC

What the set-up looks like, and what is inside the column

Photos reproduced from the CM3191 Organic Lab Manual AY26/27 (Figures 5 and 6, pp. 25–26). Schematic not to scale.



Manual Figure 5: a silica gel column

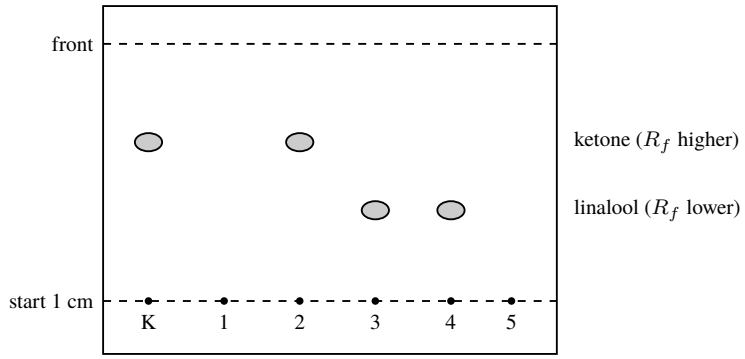


Inside the column



Manual Figure 6: TLC, $KMnO_4$ stain

Fraction TLC plate: a sensible layout [E]



K = starting ketone reference; 1–5 = tube numbers. Spot positions are illustrative only; use your own Week-9 R_f values. Tubes showing only the lower spot are combined.

COLLECT TODAY FOR THE VIDEO REPORT (group assignment guideline)

The appendix must contain: **(1)** a fully processed ^{1}H NMR spectrum, calibrated to residual $CHCl_3$ at **7.26 ppm**, plotted from **10.0 to −0.5 ppm**, with integrals and peak picking; **(2)** a fully processed IR spectrum with major bands labelled; **(3)** scanned raw datasheets. The talk covers mechanism, spectral evidence (^{1}H NMR, IR, **TLC**) and purity/isolated yield — so photograph the fraction TLC plates and note the IR and NMR file numbers today.

Name: _____

Lab Group_____ Fumehood _____ Date of the experiment: _____

Datasheet for Synthesis of (±) Linalool using Grignard Reaction

Instruction: Take photos of the measurements. After completing the experiment, show the photos to your teaching assistant and ask them to sign your printed datasheet (hardcopy, not softcopy). Submit the signed datasheet along with your lab report

Starting materials:

- Amount of 6-methyl-5-hepten-2-one used (mL)_____
- Amount of vinyl magnesium bromide used (mL)_____

Crude product

- Mass of the empty rbf (g)_____
- Mass of product + rbf (g) _____
- Mass of your final product (g) _____

Purified product

- Mass of the empty rbf (g)_____
- Mass of product + rbf (g) _____
- Mass of your final product (g) _____

Observations

Draw your TLC plate here and label all distances (in cm)